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MODULE 1: CELL BIOLOGY

1. (Multiple Choice) Which of the following organelles is responsible for the degradation of very long-chain fatty acids through beta-oxidation, producing hydrogen peroxide as a byproduct?
 - A. Lysosome
 - B. Smooth Endoplasmic Reticulum
 - C. Peroxisome
 - D. Mitochondrion
 - E. Golgi Apparatus
2. (True or False) During DNA replication in eukaryotes, the lagging strand is synthesized continuously in the 5' to 3' direction.
3. (Open Question) Explain the function of the G1/S cell cycle checkpoint (Restriction Point). Mention the role of the Retinoblastoma (Rb) protein and E2F transcription factors.
4. (Fill in the Gap) In the nucleolus, the 45S precursor rRNA is synthesized by RNA Polymerase _____.
5. (Matching) Match the cytoskeletal filament with its key characteristic or protein subunit.
6. Microtubules
7. Actin filaments (Microfilaments)
8. Intermediate filaments
9. Nuclear lamins
 - A. Composed of tubulin heterodimers; originate from the centrosome.
 - B. Provide strong mechanical and tensile strength to epithelial cells (e.g., keratins).
 - C. Highly concentrated beneath the plasma membrane; involved in cell crawling.
 - D. Form a supportive meshwork on the inner face of the nuclear envelope.
10. (Multiple Choice) Which general transcription factor possesses helicase activity to unwind DNA and kinase activity to phosphorylate the C-terminal domain of RNA Polymerase II?
 - A. TFIID
 - B. TFIIIB
 - C. TFIIF
 - D. TFIIE
 - E. TFIIA
11. (True or False Cluster) Evaluate the following statements regarding G-Protein Coupled Receptors (GPCRs):
 - A. They contain exactly seven transmembrane alpha-helical domains.
 - B. Upon ligand binding, the alpha subunit of the G-protein exchanges GDP for GTP.
 - C. The activated G-protein complex directly phosphorylates target gene promoters in the nucleus.
 - D. Adenylyl cyclase is a common effector enzyme activated by G-alpha-s (stimulatory).
12. (Open Question) Describe the molecular mechanism of crossing-over (homologous recombination). During which specific phase of meiosis does it occur, and what is its primary biological advantage?
13. (Multiple Choice) Most mitochondrial proteins are encoded by the nuclear genome. What specific sequence directs a fully translated, unfolded cytosolic protein to the mitochondria?
 - A. C-terminal KDEL sequence
 - B. N-terminal amphipathic alpha-helix
 - C. Mannose-6-phosphate tag
 - D. Proline-rich nuclear localization signal (NLS)
 - E. Signal recognition particle (SRP) binding motif
14. (Fill in the Gap) The carbohydrate-rich coat on the extracellular surface of the plasma membrane, important for cell recognition and protection, is called the _____.

15. (True or False) DNA methylation typically occurs at cytosine residues within CpG islands and is generally associated with transcriptional repression (gene silencing).
16. (Multiple Choice) In the regulation of apoptosis, which of the following proteins acts as an anti-apoptotic guardian by maintaining mitochondrial membrane integrity?
- A. Bax
 - B. Bak
 - C. Bcl-2
 - D. Cytochrome c
 - E. Caspase-3
17. (Matching) Match the DNA repair mechanism with the type of damage it corrects.
18. Base Excision Repair (BER)
19. Nucleotide Excision Repair (NER)
20. Non-Homologous End Joining (NHEJ)
21. Mismatch Repair (MMR)
- A. Fixes bulky, helix-distorting lesions like thymine dimers caused by UV light.
 - B. Repairs double-strand breaks but often introduces small insertions/deletions.
 - C. Removes single oxidized, deaminated, or alkylated bases.
 - D. Corrects uncorrected replication errors right after the replication fork passes.
22. (Open Question) Describe the role of SNARE proteins and Rab GTPases in the process of vesicular transport and membrane fusion.
23. (Multiple Choice) Embryonic stem cells (ESCs) derived from the inner cell mass are considered:
- A. Totipotent
 - B. Pluripotent
 - C. Multipotent
 - D. Unipotent
 - E. Nullipotent
24. (True or False Cluster) Evaluate the following statements regarding telomeres and telomerase:
- A. Telomeres consist of long, repetitive, non-coding DNA sequences.
 - B. Telomerase is a DNA-dependent RNA polymerase.
 - C. Telomerase activity is typically high in somatic cells and absent in cancer cells.
 - D. Telomere shortening acts as a biological clock, eventually triggering replicative senescence.
25. (Fill in the Gap) Proteins targeted for degradation by the 26S proteasome are covalently tagged with a poly-_____ chain.
26. (True or False) Retroviruses like HIV use the enzyme integrase to insert their newly synthesized double-stranded DNA into the host cell's genome.
27. (Open Question) Distinguish between polygenic inheritance and pleiotropy, providing a brief conceptual definition of each.
28. (Multiple Choice) The Ras protein is frequently mutated in human cancers. In its normal proto-oncogene state, Ras functions as a:
- A. Transmembrane tyrosine kinase receptor
 - B. Monomeric G-protein (small GTPase)
 - C. Nuclear transcription factor
 - D. Cell cycle checkpoint kinase
 - E. Caspase activator
29. (Fill in the Gap) The process by which cells engulf large particles, such as bacteria or cellular debris, into large vesicles for lysosomal degradation is called _____.
30. (Matching) Match the signaling receptor class to its mechanism of action.
31. Receptor Tyrosine Kinase (RTK)
32. GPCR
33. Ion-Channel-Coupled Receptor

34. Nuclear Receptor
- A. Opens a pore to allow specific ions to cross the membrane upon ligand binding.
 - B. Dimerizes and autophosphorylates upon ligand binding.
 - C. Acts as a transcription factor in the nucleus when bound to a lipophilic ligand.
 - D. Activates a heterotrimeric GTP-binding protein to initiate a signaling cascade.
35. (True or False) The addition of a 5' cap and a 3' poly-A tail to eukaryotic pre-mRNA occurs in the cytoplasm immediately prior to translation.
36. (Multiple Choice) During mitosis, microtubules of the mitotic spindle attach to chromosomes at specialized protein complexes called:
- A. Centrosomes
 - B. Centrioles
 - C. Kinetochores
 - D. Telomeres
 - E. Nucleosomes
37. (Open Question) Explain the cellular function of molecular chaperones (e.g., HSP70 family). Why is their activity often upregulated during conditions of cellular stress, such as elevated temperatures?
38. (Fill in the Gap) A visual representation of an individual's complete set of metaphase chromosomes, arranged in homologous pairs from largest to smallest, is called a _____.
39. (True or False Cluster) Evaluate the following statements regarding translation:
- A. The start codon AUG establishes the reading frame and codes for Methionine.
 - B. The ribosome reads the mRNA template in the 3' to 5' direction.
 - C. Peptide bond formation is catalyzed by the rRNA of the large ribosomal subunit.
 - D. Termination occurs when a release factor binds to a stop codon in the A site.
40. (Open Question) Define the concept of a tumor suppressor gene using p53 as an example. Explain why mutations in tumor suppressor genes are generally considered genetically recessive at the cellular level.
41. (Multiple Choice) In the CRISPR-Cas9 genome editing system, when Cas9 induces a double-strand break and a homologous repair template is explicitly provided, the cell will likely repair the break via:
- A. Non-Homologous End Joining (NHEJ)
 - B. Homology-Directed Repair (HDR)
 - C. Base Excision Repair
 - D. Nucleotide Excision Repair
42. (Open Question) Briefly describe the intracellular signaling pathway triggered by the activation of Phospholipase C (PLC). Identify the two secondary messengers generated and their immediate cellular targets.

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MODULE 2: HISTOLOGY

31. (Multiple Choice) Which type of epithelium, characterized by dome-shaped superficial "umbrella" cells, lines the renal pelvis, ureters, and urinary bladder?
- A. Stratified squamous non-keratinized epithelium
 - B. Pseudostratified ciliated columnar epithelium
 - C. Transitional epithelium (Urothelium)
 - D. Simple columnar epithelium
32. (Open Question) Compare the histological structure and secretory delivery mechanisms of exocrine glands versus endocrine glands.
33. (True or False) Mast cells contain large basophilic granules filled with histamine and heparin, playing a key role in allergic and inflammatory responses.
34. (Fill in the Gap) The fundamental structural unit of compact bone, consisting of concentric lamellae of bone matrix surrounding a central canal, is the _____ (or Haversian system).
35. (Matching) Match the mature leukocyte to its primary histological description or function.
36. Neutrophil

37. Eosinophil
38. Basophil
39. Monocyte
- A. Large, kidney-shaped nucleus; differentiates into macrophages in tissues.
 - B. Multilobed nucleus; first responder to bacterial infections; highly phagocytic.
 - C. Bilobed nucleus; large red-staining granules; combats parasitic infections.
 - D. Obscured nucleus due to large purple-staining granules; mediates inflammation.
40. (Multiple Choice) Which type of cartilage lacks a distinct perichondrium and features parallel bundles of Type I collagen, making it highly resistant to compression and shear forces (e.g., in intervertebral discs)?
- A. Hyaline cartilage
 - B. Elastic cartilage
 - C. Fibrocartilage
 - D. Articular cartilage
41. (Open Question) Describe the histological structure of a skeletal muscle sarcomere. Explain what happens to the I band, A band, and H zone during muscle contraction according to the sliding filament model.
42. (True or False Cluster) Evaluate the following statements regarding nervous tissue glia:
- A. Astrocytes contribute to the formation and maintenance of the blood-brain barrier.
 - B. Microglia are derived from the neuroectoderm.
 - C. Ependymal cells line the ventricles of the brain and produce cerebrospinal fluid.
 - D. Oligodendrocytes can myelinate multiple axons simultaneously in the CNS.
43. (Multiple Choice) Which of the following lymphoid organs is characterized histologically by the presence of a cortex and a medulla, lobules separated by connective tissue septa, and unique structures called Hassall's corpuscles?
- A. Spleen
 - B. Lymph node
 - C. Palatine tonsil
 - D. Thymus
44. (Fill in the Gap) The acellular supportive structure secreted by epithelial cells, composed primarily of laminin, Type IV collagen, and entactin/nidogen, is the _____ lamina.
45. (Multiple Choice) White adipose tissue is histologically classified as:
- A. Multilocular
 - B. Unilocular
 - C. Non-vascularized
 - D. Chondrogenic
46. (True or False) Smooth muscle cells lack T-tubules; instead, they possess numerous flask-shaped plasma membrane invaginations called caveolae that help sequester calcium.
47. (Open Question) Briefly distinguish between intramembranous ossification and endochondral ossification. Name one specific bone in the human body that develops via each mechanism.
48. (Fill in the Gap) In the peripheral nervous system, small islands of Schwann cell cytoplasm trapped within the myelin sheath are visible histologically as _____ clefts.
49. (Matching) Match the intercellular junction to its primary transmembrane protein(s).
50. Tight Junctions (Zonula Occludens)
51. Adherens Junctions (Zonula Adherens)
52. Desmosomes (Macula Adherens)
53. Gap Junctions
- A. Connexins
 - B. Desmoglein and Desmocollin (Cadherin family)
 - C. E-cadherin
 - D. Claudins and Occludins
54. (Open Question) Name the layers of the thick epidermis in order from deep to superficial, briefly describing the defining morphological feature of the stratum granulosum.

55. (True or False) In the spleen, the red pulp functions primarily to filter and degrade senescent erythrocytes, whereas the white pulp is lymphoid tissue involved in adaptive immune responses.
56. (Multiple Choice) In skeletal muscle, a “triad” is located at the A-I band junction and is composed of:
- A. Two T-tubules and one terminal cisterna of the sarcoplasmic reticulum.
 - B. One T-tubule and two terminal cisternae of the sarcoplasmic reticulum.
 - C. Three aligned T-tubules.
 - D. One T-tubule, one mitochondrion, and one Z-line.
57. (Fill in the Gap) The most abundant cell type in proper connective tissue, responsible for synthesizing collagen, elastin, and ground substance, is the _____.
58. (Open Question) Outline the general sequence of erythropoiesis in the bone marrow (from proerythroblast to mature erythrocyte). At what stage is the nucleus extruded?
59. (Multiple Choice) Large, flask-shaped neurons with highly branched dendritic trees that are uniquely found in the cortex of the cerebellum are called:
- A. Pyramidal cells
 - B. Purkinje cells
 - C. Bipolar cells
 - D. Anterior horn cells
60. (True or False Cluster) Evaluate the following statements regarding bone remodeling:
- A. Osteoclasts secrete hydrochloric acid and lysosomal enzymes (e.g., cathepsin K) to degrade matrix.
 - B. Osteoblasts differentiate from the hematopoietic stem cell lineage.
 - C. Parathyroid hormone (PTH) directly binds to receptors on osteoclasts to stimulate them.
 - D. Osteocytes reside in lacunae and act as mechanosensors.
61. (Multiple Choice) A gland that releases its secretory product by pinching off the apical portion of the cell cytoplasm (e.g., lactating mammary gland) utilizes which mode of secretion?
- A. Merocrine
 - B. Holocrine
 - C. Apocrine
 - D. Endocrine
62. (Open Question) Compare the process of axon myelination in the Central Nervous System (CNS) versus the Peripheral Nervous System (PNS), specifically identifying the glial cells involved.

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MODULE 3: EMBRYOLOGY

55. (Multiple Choice) Following fertilization, the zygote undergoes rapid mitotic divisions called cleavage without an increase in overall cell mass. This process leads to the formation of a solid ball of 16-32 cells known as the:
- A. Blastocyst
 - B. Morula
 - C. Trophoblast
 - D. Epiblast
 - E. Hypoblast
56. (True or False) The notochord develops from the endoderm layer and persists into adulthood as the entire spinal cord.
57. (Open Question) Compare spermatogenesis and oogenesis regarding the timing of meiosis onset and the ultimate yield of functional gametes from a single primary germ cell.
58. (Fill in the Gap) During the second week of development, the trophoblast differentiates into two layers: the inner cytotrophoblast and the outer, highly invasive _____.
59. (Matching) Match the embryonic germ layer or structure to its adult derivative.
60. Ectoderm
61. Mesoderm (Paraxial)

62. Endoderm
63. Neural Crest
- A. Epithelial lining of the respiratory tract.
 - B. Epidermis of the skin and the central nervous system.
 - C. Melanocytes and facial cartilage.
 - D. Somites (skeletal muscle, vertebrae, dermis).
64. (Multiple Choice) A normal, full-term umbilical cord typically contains which of the following blood vessels?
- A. Two umbilical veins (carrying oxygenated blood) and one umbilical artery.
 - B. Two umbilical arteries (carrying deoxygenated blood) and one umbilical vein.
 - C. One umbilical artery and one umbilical vein.
 - D. Two umbilical arteries and two umbilical veins.
65. (Open Question) Describe the process of gastrulation. Include the role of the primitive streak and identify the three germ layers that are ultimately formed from the epiblast.
66. (True or False Cluster) Evaluate the following statements regarding fertilization:
- A. Capacitation is a biochemical conditioning process sperm must undergo in the female reproductive tract.
 - B. The acrosome reaction involves the release of enzymes to penetrate the corona radiata only.
 - C. The zona reaction prevents polyspermy by altering the zona pellucida glycoproteins.
 - D. Fertilization most commonly occurs in the ampulla of the uterine tube.
67. (Multiple Choice) Failure of the caudal neuropore to close properly during the fourth week of development results in:
- A. Anencephaly
 - B. Spina bifida
 - C. Gastroschisis
 - D. Ectopia cordis
68. (True or False) The pharyngeal (branchial) arches are composed of a core of mesenchyme covered externally by ectoderm and internally by endoderm, contributing heavily to the formation of the head and neck.
69. (Multiple Choice) The extraembryonic membrane that surrounds the embryo in a fluid-filled sac, providing protection against mechanical trauma and preventing adhesion to surrounding tissues, is the:
- A. Yolk sac
 - B. Allantois
 - C. Chorion
 - D. Amnion
70. (Fill in the Gap) In female gametogenesis, the secondary oocyte halts its meiotic division at _____ II, completing it only if fertilization occurs.

MODULE 4: INTERDISCIPLINARY

67. (Multiple Choice) Lysosomal storage diseases (e.g., Tay-Sachs) result from the genetic deficiency of specific lysosomal hydrolases. Histologically, which tissue type is typically most severely affected, displaying massive intracellular accumulation of undegraded substrates that impairs function and leads to early death?
- A. Stratified squamous epithelium
 - B. Connective tissue (adipocytes)
 - C. Nervous tissue (neurons)
 - D. Skeletal muscle tissue
68. (True or False) Epidermolysis Bullosa is caused by a genetic defect in hemidesmosomes or Type VII collagen, resulting in a histological failure of the epidermis to adhere to the underlying dermis.
69. (Open Question) Duchenne Muscular Dystrophy (DMD) is caused by a mutation in the dystrophin gene. Integrating cell biology and histology, explain the normal function of the dystrophin protein complex and describe the histological appearance of a skeletal muscle biopsy from a DMD patient over time.
70. (Fill in the Gap) Sonic Hedgehog (SHH) is a critical morphogen secreted by the Zone of Polarizing Activity (ZPA) during embryonic limb development. It primarily regulates the _____-posterior patterning of the developing limb.

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ANSWER KEY & RUBRIC

1. C (Peroxisomes handle beta-oxidation of very long-chain fatty acids)
2. False (The lagging strand is synthesized discontinuously via Okazaki fragments; synthesis is always 5' to 3' locally, but moves 3' to 5' relative to the fork)
3. [Rubric - OQ]: The G1/S checkpoint ensures the cell is ready for DNA replication. Growth factors activate CDKs that phosphorylate Retinoblastoma (Rb). Phosphorylated Rb releases E2F transcription factors, which then enter the nucleus to transcribe S-phase genes.
4. I (RNA Polymerase I)
5. 1-A, 2-C, 3-B, 4-D
6. D (TFIIH unwinds the promoter DNA and phosphorylates the RNA Pol II tail to begin transcription)
7. A-True, B-True, C-False, D-True
8. [Rubric - OQ]: Crossing-over is the physical exchange of genetic material between non-sister chromatids of homologous chromosomes. It occurs during Prophase I of meiosis (pachytene stage) and increases genetic variability in offspring.
9. B (An N-terminal amphipathic alpha-helix directs proteins to mitochondrial import receptors like TOM/TIM)
10. Glycocalyx
11. True (DNA methylation at CpG islands represses transcription)
12. C (Bcl-2 is anti-apoptotic; Bax and Bak are pro-apoptotic pore formers)
13. 1-C, 2-A, 3-B, 4-D
14. [Rubric - OQ]: Rab GTPases regulate initial vesicle docking and tethering to target membranes. SNARE proteins (v-SNAREs on the vesicle and t-SNAREs on the target) intertwine to pull the membranes closely together, providing the mechanical force necessary for lipid bilayer fusion.
15. B (Pluripotent - they can form all three germ layers but not extraembryonic trophoblast)
16. A-True, B-False, C-False, D-True
17. Ubiquitin
18. True (Integrase inserts viral DNA into host chromatin)
19. [Rubric - OQ]: Polygenic inheritance occurs when multiple genes combine to determine a single phenotypic trait (e.g., height). Pleiotropy occurs when a single gene mutation results in multiple, distinct, and seemingly unrelated phenotypic traits (e.g., Marfan syndrome).
20. B (Ras is a monomeric small GTPase that relays signals from RTKs to the MAPK cascade)
21. Phagocytosis
22. 1-B, 2-D, 3-A, 4-C
23. False (Capping and polyadenylation occur in the nucleus, not the cytoplasm)
24. C (Kinetochore assembles at the centromere)
25. [Rubric - OQ]: Chaperones prevent protein aggregation and assist in proper protein folding by binding to exposed hydrophobic regions. Their expression increases during heat stress (Heat Shock Proteins) because heat denatures proteins, increasing the need for refolding assistance.
26. Karyotype
27. A-True, B-False, C-True, D-True
28. [Rubric - OQ]: Tumor suppressor genes normally inhibit cell division or promote apoptosis (e.g., p53 arrests the cycle for DNA repair). They are recessive at the cellular level because a single healthy allele usually produces enough functional protein to maintain control; both alleles must be mutated (loss of function) for cancer to progress.
29. B (HDR uses the provided template for precise repair; NHEJ is error-prone and joins ends without a template)
30. [Rubric - OQ]: PLC cleaves PIP₂ into Inositol triphosphate (IP₃) and Diacylglycerol (DAG). IP₃ triggers calcium release from the ER. DAG and calcium together activate Protein Kinase C (PKC).
31. C (Transitional epithelium / Urothelium)
32. [Rubric - OQ]: Exocrine glands retain a connection to the surface epithelium via a duct system, secreting their products into the duct. Endocrine glands lose their ductal connection and secrete their products (hormones) directly into the surrounding rich capillary network (bloodstream).
33. True
34. Osteon
35. 1-B, 2-C, 3-D, 4-A
36. C (Fibrocartilage)
37. [Rubric - OQ]: A sarcomere extends from Z-line to Z-line. During contraction, the thin actin filaments slide over the thick myosin filaments. The A band (myosin length) remains constant. The I band (only actin) and H zone (only myosin) shorten or disappear.
38. A-True, B-False, C-True, D-False
39. D (Thymus)
40. Basal (Basal lamina)

41. B (Unilocular - containing a single large lipid droplet)
42. True (Caveolae serve a functionally similar role to T-tubules in smooth muscle)
43. [Rubric - OQ]: Intramembranous ossification occurs directly from mesenchymal tissue without a cartilage intermediate (e.g., flat bones of the skull). Endochondral ossification replaces a pre-existing hyaline cartilage model with bone (e.g., long bones like the femur).
44. Schmidt-Lanterman
45. 1-D, 2-C, 3-B, 4-A
46. [Rubric - OQ]: Basale, Spinosum, Granulosum, Lucidum (thick skin only), Corneum. The stratum granulosum is defined by the presence of dense, basophilic keratohyalin granules involved in keratinization and barrier formation.
47. True
48. B (One T-tubule flanked by two terminal cisternae of the SR)
49. Fibroblast
50. [Rubric - OQ]: Proerythroblast -> Basophilic erythroblast -> Polychromatophilic erythroblast -> Orthochromatophilic erythroblast (normoblast) -> Reticulocyte -> Erythrocyte. The nucleus is extruded at the orthochromatophilic stage to form the reticulocyte.
51. B (Purkinje cells)
52. A-True, B-False, C-False, D-True
53. C (Apocrine)
54. [Rubric - OQ]: In the PNS, a single Schwann cell wraps its membrane around a single axon segment to form myelin. In the CNS, a single Oligodendrocyte extends multiple processes to simultaneously myelinate multiple separate axon segments.
55. B (Morula)
56. False (It develops from mesoderm/epiblast and eventually forms the nucleus pulposus of intervertebral discs, not the spinal cord)
57. [Rubric - OQ]: Spermatogenesis begins at puberty, is continuous, and yields 4 functional spermatozoa per primary spermatocyte. Oogenesis begins in fetal life, arrests until puberty, yields only 1 functional ovum (and polar bodies) per primary oocyte, and ends at menopause.
58. Syncytiotrophoblast
59. 1-B, 2-D, 3-A, 4-C
60. B (Two umbilical arteries carrying deoxygenated blood to the placenta, one umbilical vein carrying oxygenated blood to the fetus)
61. [Rubric - OQ]: Gastrulation begins with the formation of the primitive streak on the epiblast surface. Epiblast cells invaginate through the streak to displace the hypoblast (forming definitive endoderm), fill the middle space (forming mesoderm), and remain on the surface (forming ectoderm).
62. A-True, B-False, C-True, D-True
63. B (Spina bifida results from caudal neuropore failure; anencephaly from cranial)
64. True
65. D (Amnion)
66. Metaphase
67. C (Neurons are highly vulnerable to undegraded lipid/ganglioside accumulation)
68. True
69. [Rubric - OQ]: Dystrophin links the intracellular actin cytoskeleton to the extracellular matrix via transmembrane proteins, providing mechanical stability. In DMD biopsies, muscle fibers vary widely in size, show centralized nuclei (attempted regeneration), and are progressively replaced by fibrotic connective tissue and fat.
70. Anterior (Anterior-posterior patterning)